Statistics/Data analysis

## Title

**lpbwdensity** —— Bandwidth Selection for Local Polynomial Density Estimation and Inference.

## Syntax

**lpbwdensity** *Var* [*if*] [*in*] [,
**grid(***Var***) p(***#***) v(***#***) kernel(***KernelFn* **)bwselect(***BwMethod***) nomasspoints nostdvar**
**nlocalmin(***#***) nuniquemin(***#***) noregularize**
**cweights(***Var***) pweights(***Var***)**
**genvars(***NewVarName***)**
**rgrid(***Var***) rindex(***Var***) separator(***#***) precision(***Precision***)**
]

## Description

**lpbwdensity** implements the bandwidth selection methods for local polynomial based density (and derivatives) estimation proposed and studied in Cattaneo, Jansson and Ma (2020) and Cattaneo, Jansson and Ma (2024). See Cattaneo, Jansson and Ma (2022) for more implementation details and illustrations.

Companion command: lpdensity for estimation and robust bias-corrected inference.

Companion R functions are also available here.

Related Stata and R packages are available in the following website:

https://nppackages.github.io/

## Options

### Bandwidth Selection

**grid**(*Var*) specifies the grid on which density is estimated. When set to default, grid points will be chosen as 0.05-0.95 percentiles of the data, with 0.05 step size.

**p**(*#*) specifies the the order of the local-polynomial used to construct point estimates. Default is **p(2)** (local quadratic regression).

**v**(*#*) specifies the derivative of distribution function to be estimated. **v(0)** for the distribution function, **v(1)** (default) for the density funtion, etc.

**kernel**(*KernelFn*) specifies the kernel function used to construct the local-polynomial estimator(s).
**triangular** *K(u) = (1 - |u|) * (|u|<=1)*. This is the default option.
**epanechnikov** *K(u) = 0.75 * (1 - u^2) * (|u|<=1)*.
**uniform** *K(u) = 0.5 * (|u|<=1)*.

**bwselect**(*BwMethod*) specifies method for data-driven bandwidth selection. This option will be ignored if **bw(***Var***)** is provided.
**mse-dpi** mean squared error optimal bandwidth for each grid point. This is the default option.
**imse-dpi** integrated mean squared error optimal bandwidth which is common for all grid points.
**mse-rot** rule-of-thumb bandwidth based on a Gaussian reference model.
**imse-rot** integrated rule-of-thumb bandwidth based on a Gaussian reference model which is common for all grid points.

**nomasspoints** will not adjust for mass points in the data.

**nostdvar** will not standardize the data for bandwidth selection. Note that this may lead to unstable performance of the numerical optimization procedure.

### Local Sample Size Checking

**nlocalmin**(*#*) specifies the minimum number of observations in each local neighborhood. This option will be ignored if set to 0, or if **noregularize** is used. The default value is **20+p(*#*)+1**.

**nuniquemin**(*#*) specifies the minimum number of unique observations in each local neighborhood. This option will be ignored if set to 0, or if **noregularize** is used. The default value is **20+p(*#*)+1**.

**noregularize** suppresses local sample size checking.

### Weights

**cweights**(*Var*) specifies weights used for counterfactual distribution construction.

**pweights**(*Var*) specifies weights used in sampling. Should be nonnegative.

### Storing and displaying results

**genvars**(*NewVarName*) specifies if new varaibles should be generated to store estimation results. If *newVarName* is provided, the following new varaibles will be generated:
*NewVarName_grid* grid points,
*NewVarName_bw* bandwidths,
*newVarName_nh* local/effective sample sizes.

**rgrid**(*var*) specifies a set of grid points to display the results. When omitted, this will be the same as **grid(*Var*)**.

**rindex**(*var*) specifies a set of indices to display the results. This option will be ignored if **rgrid(*Var*)** is provided.

**separator**(*#*) draw a seperation line after every # variables; default is **separator(5)**.

**precision**(*Precision*) controls the storage type for command-generated temporary and **genvars()** variables. Options are **double** (default) and **single**. Mata calculations and returned matrices are double precision; for full double precision, user-supplied data, grid, bandwidth, and weight variables should also be stored as **double**.

## Examples

Generate artifitial data:

```
. set obs 1000
. set seed 42
. gen lpd_data = rnormal()
```

MSE-optimal bandwidths for empirical quantiles:

```
. lpbwdensity lpd_data
```

Save estimation results to variables:

```
. capture drop temp_*
. lpbwdensity lpd_data, genvars(temp)
```

## Saved results

**lpbwdensity** saves the following in **e()**:

Scalars

| | |
|---|---|
| **e(N)** | sample size |
| **e(p)** | option **p(#)** |
| **e(v)** | option **v(#)** |

Macros

| | |
|---|---|
| **e(bwselect)** | option **bwselect(***BwMethod***)** |
| **e(kernel)** | option **kernel(***KernelFn***)** |
| **e(precision)** | option **precision(***Precision***)** |

Matrices

| | |
|---|---|
| **e(result)** | estimation result |

## References

Cattaneo, M. D., Michael Jansson, and Xinwei Ma. 2020. Simple Local Polynomial Density Estimators.
*Journal of the American Statistical Association* 115(531): 1449-1455.

Cattaneo, M. D., Michael Jansson, and Xinwei Ma. 2022. lpdensity: Local Polynomial Density Estimation and Inference.
*Journal of Statistical Software* 101(2): 1-25.

Cattaneo, M. D., Michael Jansson, and Xinwei Ma. 2024. Local Regression Distribution Estimators.
*Journal of Econometrics* 240(2): 105074.

## Authors

Matias D. Cattaneo, Princeton University, Princeton, NJ.
matias.d.cattaneo@gmail.com.

Michael Jansson, University of California Berkeley, Berkeley, CA.
michael.jansson.berkeley@gmail.com.

Xinwei Ma, University of California San Diego, La Jolla, CA.
xinweima.pku@gmail.com.